

Learning the metric induced by a G_2 -structure

Generalisation and equivariance through data augmentation

Eduardo Toffolo

University of Campinas

MM845 — AI for Geometry

Research question

A positive 3-form φ on an oriented 7-dimensional vector space determines a unique metric g_φ .

$$\varphi_0 = e^{123} + e^{145} + e^{167} + e^{246} - e^{257} - e^{347} - e^{356}.$$

Supervised task

Learn, from the 35 coefficients of φ , the map

$$F : \mathcal{P} \subset \Lambda^3(\mathbb{R}^7)^* \longrightarrow \text{Sym}^+(7), \quad F(\varphi) = g_\varphi,$$

whose output has 28 independent entries.

Experimental question

Can a generic neural network approximate this map? Does rotation augmentation improve accuracy, generalisation and compatibility with the geometry?

The exact formula and geometric identities

Metric determined by φ

In an oriented basis with reference volume vol_0 ,

$$(B_\varphi)_{ij} \text{vol}_0 = \frac{1}{6} (\iota_{e_i} \varphi) \wedge (\iota_{e_j} \varphi) \wedge \varphi,$$
$$g_\varphi = (\det B_\varphi)^{-1/9} B_\varphi.$$

If $\varphi = A^* \varphi_0$, then

$$g_\varphi = A^T A.$$

Mathematical references: Karigiannis (2020) and Grigorian (2016).

Naturality

For $C \in \text{GL}^+(7)$,

$$F(C^* \varphi) = C^T F(\varphi) C.$$

Homogeneity

For $t > 0$,

$$F(t\varphi) = t^{2/3} F(\varphi).$$

These identities provide separate tests of equivariance and scaling.

Synthetic data and experimental protocol

Exact-label generation

$$A = Q_1 \operatorname{diag}(e^{s_1}, \dots, e^{s_7}) Q_2,$$
$$\varphi = A^* \varphi_0, \quad g = A^T A,$$

with independent Haar $Q_1, Q_2 \in \operatorname{SO}(7)$.

- ID: 30,000 sources, $s_i \in [-0.35, 0.35]$;
- OOD: 4,500 sources, $s_i \in [-0.7, 0.7]$, with at least one $|s_i| > 0.35$.

Training and evaluation

- ID split: 21,000 / 4,500 / 4,500;
- training subsets: 2,100, 7,000 and 21,000;
- seeds 11, 22 and 33: 27 fitted models;
- relative Frobenius error on ID and OOD;
- $\operatorname{SO}(7)$, $\operatorname{GL}^+(7)$, scaling and positivity tests.

Reference check

The two exact formulas agree within 6×10^{-16} ; the split precedes augmentation.

Three models under the same protocol

The training objective is the mean squared relative Frobenius error:

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{n=1}^N \frac{\|\hat{g}_{\theta}(\varphi_n) - g_n\|_F^2}{\|g_n\|_F^2}.$$

Ridge

Affine map from 35 coefficients to 28 symmetric entries. Positive definiteness is not guaranteed.

MLP

Architecture $35 \rightarrow 128 \rightarrow 128 \rightarrow 28$, with ReLU and Adam. The output forms a triangular L , with a soft plus diagonal, and $\hat{g} = LL^T$.

MLP + rotations

Same network. At each epoch and for each source,

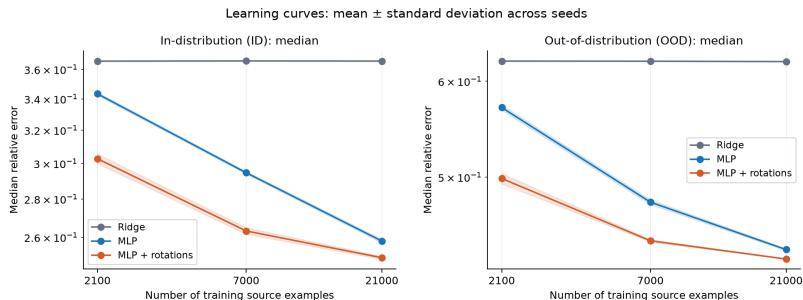
$$(\varphi, g) \mapsto (C^* \varphi, C^T g C),$$

with a new $C \in \text{SO}(7)$.

Controlled comparison

Same sources, initialisation and minibatches: augmentation changes the views but retains 21,000 examples per epoch.

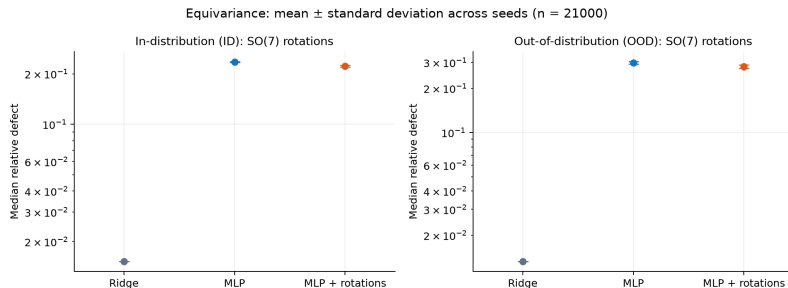
Accuracy: more data help, but the error remains high



	Ridge	MLP	MLP + rotations
ID	36.58%	25.81%	24.99%
OOD	62.29%	43.57%	42.79%

Interpretation. For 21,000 sources, these are means across three seed-level medians. The MLP outperforms ridge and augmentation helps modestly; the ID–OOD gap remains large.

Equivariance: a modest improvement and a pitfall



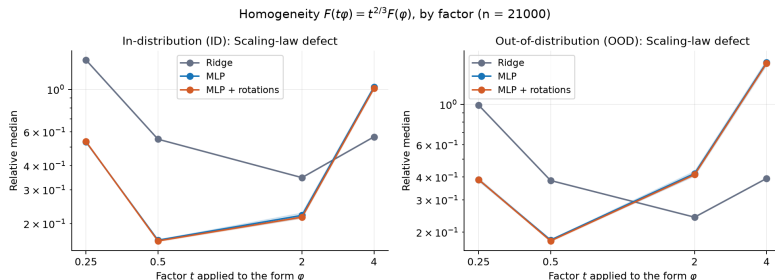
Effect of augmentation

The median SO(7) defect falls from 0.2355 to 0.2223 on ID and from 0.2985 to 0.2818 on OOD: a relative reduction of about 5.6%.

Why does ridge look better?

It almost always predicts 1.08288/. An almost constant scalar output has a low rotation defect, yet can ignore the input and predict the wrong metric.

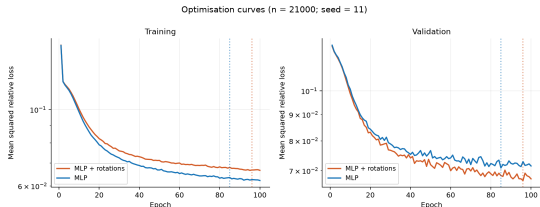
Rotation augmentation does not learn the scaling law



Result

Augmentation by $SO(7)$ does not enforce $\hat{F}(t\varphi) = t^{2/3}\hat{F}(\varphi)$. The defect grows for factors far from the training range, especially at $t = 4$, on both ID and OOD. Accuracy, equivariance and homogeneity are distinct diagnostics.

Why is the error still large?



Training and validation curves for $n = 21,000$, seed 11.

Hypotheses consistent with the results

- cubic map with determinant normalisation;
- generic MLP with no built-in geometric identities;
- best checkpoints near 100 epochs; OOD increases scale and anisotropy.

Next experiments

Test overfitting on a small sample; then increase capacity and budget, or impose homogeneity and equivariance.

Conclusions

- 1 Synthetic generation provides exact labels and supports controlled tests of geometric properties.
- 2 The MLP clearly reduces the error relative to ridge; rotation augmentation improves accuracy and the $SO(7)$ defect for every seed, but only modestly.
- 3 Median errors of approximately 25% on ID and 43% on OOD do not constitute a high-accuracy approximation.
- 4 The network does not automatically learn exact equivariance or the scaling law. The geometric formula remains the reference.

Answer to the experimental question

Data augmentation helps consistently under this protocol, but it does not replace an architecture that incorporates the mathematical structure of the problem.

Results are reproducible across three seeds; hyperparameters and checkpoints were selected on validation data.

Thank you

Questions?